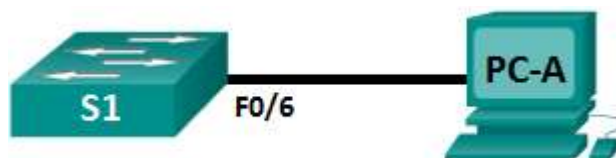


Lab2 – Viewing Network Device MAC Addresses

Chapter 5 Activity

Topology



Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway
S1	VLAN 1	192.168.1.2	255.255.255.0	N/A
PC-A	NIC	192.168.1.3	255.255.255.0	192.168.1.1

Objectives

Part 1: Configure Devices and Verify Connectivity

Part 2: Display, Describe, and Analyze Ethernet MAC Addresses

Background / Scenario

Every device on an Ethernet LAN is identified by a Layer 2 MAC address. This address is assigned by the manufacturer and stored in the firmware of the NIC. This lab will explore and analyze the components that make up a MAC address, and how you can find this information on a switch and a PC.

You will cable the equipment as shown in the topology. You will configure the switch and PC to match the addressing table. You will verify your configurations by testing for network connectivity.

After the devices have been configured and network connectivity has been verified, you will use various commands to retrieve information from the devices to answer questions about your network equipment.

Note: The switches used are Cisco Catalyst 2960s with Cisco IOS Release 15.0(2) (lanbasek9 image). Other switches and Cisco IOS versions can be used. Depending on the model and Cisco IOS version, the commands available and the output produced might vary from what is shown in the labs.

Note: Make sure that the switches have been erased and have no startup configurations. If you are unsure, ask your instructor.

Required Resources

- 1 Switch (Cisco 2960 with Cisco IOS Release 15.0(2) lanbasek9 image or comparable)
- 1 PC (Windows 7 or 8 with a terminal emulation program, such as Tera Term)
- Console cable to configure the Cisco switch via the console ports
- Ethernet cables as shown in the topology

Part 1: Configure Devices and Verify Connectivity

In this part, you will set up the network topology and configure basic settings, such as the interface IP addresses and device name. For device name and address information, refer to the Topology and Addressing Table.

Step 1: Cable the network as shown in the topology.

- Attach the devices shown in the topology and cable as necessary.
- Power on all the devices in the topology.

Step 2: Configure the IPv4 address for the PC.

- Configure the IPv4 address, subnet mask, and default gateway address for PC-A.
- From the command prompt on PC-A, ping the switch address.

Were the pings successful? Explain.

Step 3: Configure basic settings for the switch.

In this step, you will configure the device name and the IP address, and disable DNS lookup on the switch.

- Console into the switch and enter global configuration mode.

```
Switch> enable
Switch# configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#
```

- Assign a hostname to the switch based on the Addressing Table.

```
Switch(config)# hostname S1
```

- Disable DNS lookup.

```
S1(config)# no ip domain-lookup
```

- Configure and enable the SVI interface for VLAN 1.

```
S1(config)# interface vlan 1
S1(config-if)# ip address 192.168.1.1 255.255.255.0
S1(config-if)# no shutdown
S1(config-if)# end
*Mar  1 00:07:59.048: %SYS-5-CONFIG_I: Configured from console by console
```

Step 4: Verify network connectivity.

Ping the switch from PC-A. Were the pings successful? _____

Part 2: Display, Describe, and Analyze Ethernet MAC Addresses

Every device on an Ethernet LAN has a MAC address that is assigned by the manufacturer and stored in the firmware of the NIC. Ethernet MAC addresses are 48-bits long. They are displayed using six sets of hexadecimal digits that are usually separated by dashes, colons, or periods. The following example shows the same MAC address using the three different notation methods:

00-05-9A-3C-78-00

00:05:9A:3C:78:00

0005.9A3C.7800

Note: MAC addresses are also called physical addresses, hardware addresses, or Ethernet hardware addresses.

You will issue commands to display the MAC addresses on a PC and a switch, and you will analyze the properties of each one.

Step 1: Analyze the MAC address for the PC-A NIC.

Before you analyze the MAC address on PC-A, look at an example from a different PC NIC. You can issue the **ipconfig /all** command to view the MAC address of your NIC. An example screen output is shown below. When using the **ipconfig /all** command, notice that MAC addresses are referred to as physical addresses. Reading the MAC address from left to right, the first six hex digits refer to the vendor (manufacturer) of this device. These first six hex digits (3 bytes) are also known as the organizationally unique identifier (OUI). This 3-byte code is assigned to the vendor by the IEEE organization. To find the manufacturer, you can use a tool like www.macvendorlookup.com or go to the IEEE web site to find the registered OUI vendor codes. The IEEE web site address for OUI information is <http://standards.ieee.org/develop/regauth/oui/public.html>. The last six digits are the NIC serial number assigned by the manufacturer.

- a. Using the output from the **ipconfig /all** command, answer the following questions.

```
Ethernet adapter Local Area Connection:

Connection-specific DNS Suffix . . . : 
Description . . . . . : Intel(R) 82577LM Gigabit Network Connection
Physical Address. . . . . : 5C-26-0A-24-2A-60
DHCP Enabled. . . . . : No
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::b875:731b:3c7b:c0b1%10(Preferred)
IPv4 Address. . . . . : 192.168.1.3(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : 192.168.1.1
DHCPv6 IAID . . . . . : 240920024
```

What is the OUI portion of the MAC address for this device?

What is the serial number portion of the MAC address for this device?

Using the example above, find the name of the vendor that manufactured this NIC.

- b. From the command prompt on PC-A, issue the **ipconfig /all** command and identify the OUI portion of the MAC address for the NIC of PC-A.
-

Identify the serial number portion of the MAC address for the NIC of PC-A.

Identify the name of the vendor that manufactured the NIC of PC-A.

Step 2: Analyze the MAC address for the S1 F0/6 interface.

You can use a variety of commands to display MAC addresses on the switch.

- a. Console into S1 and use the **show interfaces vlan 1** command to find the MAC address information. A sample is shown below. Use output generated by your switch to answer the questions.

```
S1# show interfaces vlan 1
Vlan1 is up, line protocol is up
```

Lab – Viewing Network Device MAC Addresses

```
Hardware is EtherSVI, address is 001b.0c6d.8f40 (bia 001b.0c6d.8f40)
Internet address is 192.168.1.1/24
MTU 1500 bytes, BW 1000000 Kbit/sec, DLY 10 usec,
    reliability 255/255, txload 1/255, rxload 1/255
Encapsulation ARPA, loopback not set
Keepalive not supported
ARP type: ARPA, ARP Timeout 04:00:00
Last input never, output 00:14:51, output hang never
Last clearing of "show interface" counters never
Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 0
Queueing strategy: fifo
Output queue: 0/40 (size/max)
5 minute input rate 0 bits/sec, 0 packets/sec
5 minute output rate 0 bits/sec, 0 packets/sec
  0 packets input, 0 bytes, 0 no buffer
    Received 0 broadcasts (0 IP multicasts)
    0 runs, 0 giants, 0 throttles
    0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored
    34 packets output, 11119 bytes, 0 underruns
    0 output errors, 2 interface resets
    0 unknown protocol drops
    0 output buffer failures, 0 output buffers swapped out
```

What is the MAC address for VLAN 1 on S1?

What is the MAC serial number for VLAN 1?

What is the OUI for VLAN 1?

Based on this OUI, what is the name of the vendor?

What does bia stand for?

Why does the output show the same MAC address twice?

- b. Another way to display the MAC address on the switch is to use the **show arp** command. Use the **show arp** command to display MAC address information. This command maps the Layer 2 address to its corresponding Layer 3 address. A sample is shown below. Use output generated by your switch to answer the questions.

S1# **show arp**

Protocol	Address	Age (min)	Hardware Addr	Type	Interface
Internet	192.168.1.1	-	001b.0c6d.8f40	ARPA	Vlan1
Internet	192.168.1.3	0	5c26.0a24.2a60	ARPA	Vlan1

What Layer 2 addresses are displayed on S1?

What Layer 3 addresses are displayed on S1?

Step 3: View the ARP table on PC

Type the command **arp-a** to visualize the arp table.

Step 4: View the MAC addresses on the switch.

Issue the **show mac address-table** command on S1. A sample is shown below. Use output generated by your switch to answer the questions.

```
S1# show mac address-table
      Mac Address Table
-----
Vlan    Mac Address      Type    Ports
----    -
All     0100.0ccc.cccc   STATIC  CPU
All     0100.0ccc.cccd   STATIC  CPU
All     0180.c200.0000   STATIC  CPU
All     0180.c200.0001   STATIC  CPU
All     0180.c200.0002   STATIC  CPU
All     0180.c200.0003   STATIC  CPU
All     0180.c200.0004   STATIC  CPU
All     0180.c200.0005   STATIC  CPU
All     0180.c200.0006   STATIC  CPU
All     0180.c200.0007   STATIC  CPU
All     0180.c200.0008   STATIC  CPU
All     0180.c200.0009   STATIC  CPU
All     0180.c200.000a   STATIC  CPU
All     0180.c200.000b   STATIC  CPU
All     0180.c200.000c   STATIC  CPU
All     0180.c200.000d   STATIC  CPU
All     0180.c200.000e   STATIC  CPU
All     0180.c200.000f   STATIC  CPU
All     0180.c200.0010   STATIC  CPU
All     ffff.ffff.ffff   STATIC  CPU
1       5c26.0a24.2a60   DYNAMIC Fa0/6
Total Mac Addresses for this criterion: 21
```

Did the switch display the MAC address of PC-A? If you answered yes, what port was it on?

Reflection

1. Can you have broadcasts at the Layer 2 level? If so, what would the MAC address be?

2. Why would you need to know the MAC address of a device?
